SEWAGE MONITORING SYSTEM

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Abstract

Sewage environments contain increasing risk factors for workers due to the presence of toxic gases and other hazardous materials found in sewer manholes. These gases, when undetected, can cause serious health issues such as poisoning and respiratory issues. Currently, there are wired monitoring systems which have been established in order to improve health and safety. Although these wired systems offer certain benefits to worker safety, they also have several major limitations, including lack of portability, high cost, and inability to adapt to the many different shapes and sizes of sewage structures and confined spaces. To address these limitations, this project will utilize a real-time wearable monitoring system that can monitor the physical and environmental variables of workers in a convenient and efficient manner. The system will utilize various types of sensors to monitor environment conditions like temperature, moisture level, level of contaminants present in the air and the presence of other harmful gases. The monitoring system will also incorporate LoRa Wireless Technology that allows long-distance communications between the wearable monitoring devices and the central control room and therefore enables continual monitoring of workers operating in environment with limited access. By continuously collecting and processing data from the environment, the device will allow workers and employers to detect gas leaks and/or hazardous working conditions before a serious accident occurs. If measurements taken by the device exceed safe operating limits, alerts will be sent to both the employee and their management in order to take corrective action quickly. The creation of a wireless sensor network will provide real-time access to data, allow for ongoing monitoring, and provide centralized control of the device with less reliance on manual observation and intervention. The device was designed with an emphasis on both affordability and portability in order to be able to use it routinely in high-risk areas such as wastewater treatment facilities. Furthermore, the design of the device considers the safety and comfort of users so they will not be distracted while completing jobs. Experimental results confirm that the wearable devices are effective at simultaneously monitoring exposure to hazardous materials and assessing workers' health while on the job. Therefore, the devices have the potential to make a considerable contribution toward the prevention of workplace injuries, the timely intervention during an emergency, and the overall safety of workers in the occupational setting. By implementing the combination of sophisticated sensing technologies, the wireless transmission of data, and real-time alerts about dangerous working conditions, the above-mentioned wearable devices present a complete solution to the challenges of working in hazardous conditions in wastewater treatment facilities.

Keywords: Sewage worker safety, Toxic gas monitoring, Wearable sensor system, Real-time monitoring, Lora wireless communication, Wireless sensor networks, Occupational health and safety

I. INTRODUCTION

Workers who work in underground sewage systems face risks that, fortunately, may be managed through appropriate planning, implementation of safety procedures, and proper training. The variety of extraction techniques that are used to extract various minerals results in increased hazards associated with underground mining. Mines that are deeper pose a greater hazard to the working person than mining in a surface location. Safety and health risks to workers involved in underground and surface sewage extraction should be a priority concern for all industries that use sewage to produce minerals. Underground sewage operations have a higher level of risk than opencast operations because of the challenges associated with ventilation and the potential for mine collapse. In addition, the machinery used for excavating and the processes used during the excavation process lead to the creation of various types of safety hazards associated with all types of sewage. Almost all modern underground and surface excavated mines have implemented various safety programs and trained their workers on the safe use of machinery and provide required training for the workers on safe working conditions. All the safety programs and training have helped to create significant improvements and changes in the safety level of workers who operate in both underground and surface excavated mines. provides approximately 70% of the total amount of energy used in India, which is why coal has played an important role in India's rapid industrialization. High-altitude work environments are challenging and unique compared to work environments at lower altitudes. They represent a very specialized area of work for workers involved in sewage management at high altitudes. High-altitude sewage worker

face extreme climatic and physiological risks associated with their work. Additionally, the activities performed by these workers expose not only miners, but also many of their fellow employees, to numerous occupational health and safety hazards.

The rapid expansion of the sewage business, which currently employs thousands of people working at or above 2000 meters (m.a.s.l.), has come about due to the increasing global demand for commodities exported from developing nations and their entry into the international commodity exchange markets. The development of mine monitoring technologies has become more advanced over time, however, as we continue to witness explosions in underground mines, mine safety remains an important issue for both society and the nation. Disasters in underground mines often result from a combination of the harsh working environment as well as the changing nature of mining conditions from day to day. Therefore, the need for the installation of mine monitoring systems is imperative for the health and safety of underground mine workers. Historically, hardwired network systems were typical of traditional mine operations and have played a significant role in promoting safe mining practices. As the size of the mined area and the vertical depth of the mine continue to grow, laneways within these operations will become increasingly filled with "blind zones," where many dangerous conditions and hazards go unnoticed. Likewise, this creates a tantamount level of uncertainty and danger for miners and all other employees who enter the mine as well as creating an increase in accidents occurring from the unknown condition within the blind zones mentioned above.

The WSN (Wireless Sensor Networks) has gained popularity globally, especially in our current times. It is a unique type of ad hoc network that operates with a number of nodes through the use of multiple hops, allowing these nodes to self-organize based on user-defined parameters over a wide geographical area, thereby providing real-time monitoring of an aspect of interest. As such, they can be employed for applications such as health care, environmental monitoring, and scientific research; additionally, military applications are numerous and expand greatly with the ability to collect vast amounts of data remotely. Typically, a WSN's node structure consists of both sensor nodes (also referred to as motes) responsible for sensing or detecting a measurement, as well as base stations (also referred to as sink nodes) that serve to manage and gather collected data from their associated sensor nodes. In addition to the previously stated application ranges being affected by numerous factors (including but not limited to: scalability of the product; hardware limitations; software applications used to connect the sensor nodes; system reliability/failure; topological arrangements of the individual sensor nodes); many other elements exist in the design of the wireless sensor networks, such as their geographical area served and overall power consumption levels.

II.RELATED WORKS

J Haxhibeqiri et al. suggested an efficient system of synchronization and scheduling to enable the development of scalable LoRaWAN Networks, which can be utilized for IoT applications. Although the LoRaWAN technology is able to support long-distance and low-power communications, the use of the ALOHA-based MAC protocol results in scalability issues for Class A devices. To resolve this, the authors proposed a single centralized scheduling component for synchronization and assignment of both uplink and downlink slots to end-devices. An efficient probabilistic data structure has been created that will help in reducing the size of synchronization packets and energy consumption associated with them, by providing a method of encoding assigned time slots. By utilizing the solution, a significant decrease in the number of packet collisions and an annual battery drain of less than 3 mAh will be experienced, which makes the solution a viable option for larger-scale deployments of IoT Systems over extended periods of time.

S Liu et al. proposed a hybrid fault diagnosis model for water quality monitoring devices that can improve the reliability of such devices for use in aquaculture. The model incorporates the use of RBDT (Rule-Based Decision Trees) in conjunction with MSVM (Multiclass Support Vector Machines). The RBDT is utilized to diagnose gateway and transmitter faults, while at the same time performing feature selection. The MSVM classifier is used to diagnose sensor faults due to the robustness and generalized capabilities of the classifier. Through these systems, S Liu et al are proposing to improve the fault diagnosis process for water quality monitoring devices used in aquaculture.

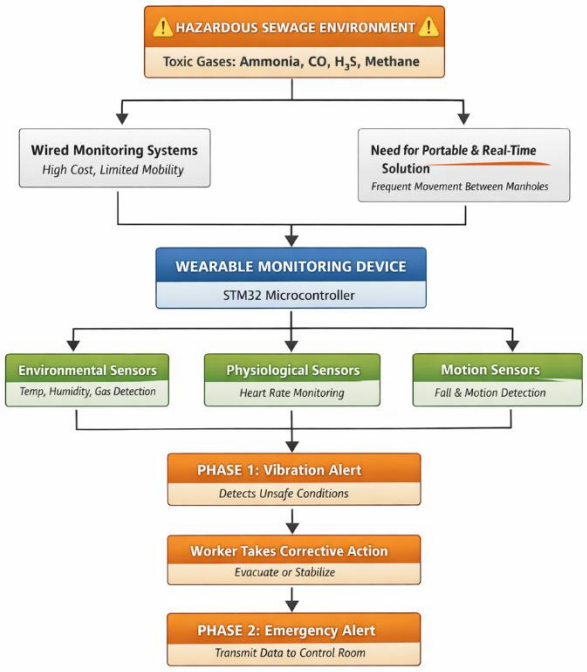
Y. Zhang et al. developed a novel method for detecting and analyzing traffic patterns in real-time using mobile GPS data collected from smart devices such as smartphones and tablets. Existing methods utilize either fixed-position cameras or other sensors placed strategically along roadways to detect vehicles and other objects on street surfaces. As a result, these methods are limited by the amount of data they can capture and the amount of time they can analyze it. To overcome these limitations, the authors propose a new approach that relies on using spatial information gleaned from mobile GPS devices to reconstruct historical traffic patterns based on time and location of mobile device use. This information is then used to develop predictive models that provide insights into future traffic patterns. Using historical data, spatial analysis techniques (e.g., clustering and geospatial analysis), and predictive modeling techniques (e.g., time-series forecasting) will improve the efficiency of current and future traffic operations. As a result, the proposed method provides an innovative approach to improve transportation operation and management through real-time analysis of navigation device use patterns. The overall goal of this work is to create better communication among transportation agencies and the public concerning traffic safety and the likelihood of creating unsafe conditions when traveling during peak traffic hours.

Q. Li et al Has Proposed a Cooperative Ports Navigation Algorithm for Autonomously Operated Underwater Vessels Based on A Student's T Extended Kalman Filter (Sekf). The Benefits of This Student's T-based Filter for Autonomous Underwater Vehicle Cooperative Navigation Include: Better Resilience to Outlier Nature of the Under Water Acoustic Communication Channel and To the Under Water Acoustic Motion Estimation Produced by Low-Cost Sensors Currently Used by These Ocean Vehicles. The Authors Developed the Sekf-Based Leader-Slave Navigation Framework and Examined How Varying Degrees Of Freedom Affect the Sekf's Roboteness. The Results Of Their Simulation And Field Trials Have Confirmed That The Proposed Method Outperforms All Other Conventional Ekf Based Methods In Accuracy And Reliability.

A. Sajid et al's Paper A Comprehensive Review Of The Security Risks Encountered With Cloud Assisted Internet Of Things Enabled Supervisory Control And Data Acquisition Systems. The Authors Illustrated How The Incorporation Of CPs Into IoT, Cloud Computing, And Wireless Sensor Networks Allows For Greater Flexibility And Fault Tolerance In Many Industrial Applications (E.G., Smart Grids, Healthcare, And Transportation). However, The Authors Also Stated That Traditional Scada Systems Are Inadequately Protected By Society And That Their Use In Combination With Newer Iot Cloud Architectures Introduces Significant Vulnerabilities. The Authors Discuss Current Limitations In SCADA Security, And Present Directions For Future Research To Successfully Implement Secure Indusive SCADA Deployments.

III.MATERIAL AND METHODS

The dangers to sewage workers are many times greater than that of non-sewage workers because sewage contains higher concentrations of hazardous materials such as ammonia; carbon monoxide; hydrogen sulfide; and methane, which can quickly become trapped in confined areas where there is normally little or no ventilation or visibility. When these gases go undetected, extended exposure may develop into significant health problems, including respiratory difficulties, lethargy, dizziness, unconsciousness, and occasionally, death. Sewage inspection and maintenance personnel are at particularly high risk because of the highly unpredictable nature of underground settings. In an effort to enhance worker safety, facilities for sewage have previously implemented wired monitoring systems. These systems offer general monitoring capabilities; but the implementation of wired systems remains challenging because of several drawbacks associated with installation and maintenance costs, as well as their required installation and continued operation in physical locations with different shapes and depths for sewage. Furthermore, hardwired systems may be physically damaged and are not able to be used successfully when sewage workers are required to frequently move between separate manholes and confined spaces, suggesting that a portable, dependable, and real-time monitoring system would provide a more dependable solution. To fill those gaps, there is an effort underway to develop a system that will allow for the ability to monitor health and safety in real time through the creation of a wearable health and safety device.



Block Diagram

The wearable unit has been built around the STM32 microcontroller as the central processing unit (CPU) to collect, process, and analyze the data gathered from various environmental sensors. Using integrated environmental sensors, it collects information regarding ambient temperature, humidity, and air quality as well as using gas sensors to monitor the presence and concentration of toxic gases typically present in sewage environments. Given the ability to measure various environmental conditions, this technology can assist in identifying unsafe working conditions and protecting workers from injury and exposure. The system continuously monitors the physiological condition of workers by recording their vital signs such as their heart rate. Knowing one's heart rate while working will indicate physical stress, exhaustion, or elevated risk of having a medical event occur during a long or arduous day of work. The system also combines motion sensors that detect rapid changes in acceleration and orientation to help identify signs of slipping or falling, losing balance while working in a constrained space. To provide a higher level of safety to the workers and to reduce the amount of false alarms, the wearable unit utilizes a two-phase mode of operation. The first phase is initiated by detecting abnormally high levels of air pollutants or unusual motion patterns within the working environment; this triggers a vibration alert through a vibration module integrated into the system to notify the worker of their potential danger. In this phase, the worker can take the appropriate corrective action, such as evacuating the area, to protect themselves and avoid potential harm.

COMPONENTS

The integrated hardware components of the suggested sewage monitoring system are intended to guarantee worker safety and efficient environmental monitoring. As the central processing unit, the Arduino Nano is in charge of gathering sensor data, carrying out control logic, and overseeing module connection. Using a transformer, rectifier, filter capacitors, and voltage regulator, a regulated power supply unit generates a steady 5V DC output, guaranteeing dependable operation of all electronic components. Methane, LPG, propane, and carbon monoxide are among the dangerous and flammable gases that are frequently found in sewage situations. Gas sensors like MQ-6 are used to identify these gases. In order to detect hazardous thermal circumstances that could endanger worker health, a temperature sensor continuously measures the surrounding air temperature. Relative humidity levels, which are crucial for determining comfort, the risk of dehydration, and general environmental safety, are monitored by the humidity sensor. Real-time sensor readings, alarms, and system status are shown locally at the manhole or control unit on a 16x2 LCD display. When the temperature, humidity, or gas concentration surpasses acceptable threshold levels, a buzzer serves as an audible alarm that promptly notifies workers. RFID modules can be used to track and identify employees, guaranteeing allowed access and enhancing personnel management. Sensor data can be transmitted from the sewage site to a distant control room or Internet of Things webpage for real-time monitoring thanks to wireless communication modules like ZigBee, LoRa, GSM, or Wi-Fi.

IV. STATISTICAL ANALYSIS

A comprehensive statistical analysis of the reliability, feasibility, and effectiveness of the proposed LoRa-based system for monitoring the health and environment of sewage workers will be conducted using the IBM SPSS Statistics Software Suite. Experimental data obtained from the wearable monitoring system that includes environmental parameters such as temperature, humidity, and levels of toxic gases; and physiological parameters such as heart rate and levels of oxygen saturation, will be converted into numerical codes and then input into SPSS for a statistical process. An initial descriptive analysis will be performed to summarize the data collected. For each parameter being monitored, descriptive statistics the mean, standard deviation, minimum, maximum, and frequency counts — will be generated so that information about both "normal" and "abnormal" working conditions can be determined in the sewage environment. These statistics will assist in identifying how stable and consistent a particular sensor performs when operated in real time. To evaluate the effectiveness of the system, inferential statistics will be employed. The Levene's Test for Equality of Variances will be applied so that the assumption of the equality of variance can be verified before the execution of parametric statistical tests. Upon completion of the Levene's test analysis, a conclusion will be made regarding whether equal variances can be assumed or not assumed for the purpose of further statistical analysis.

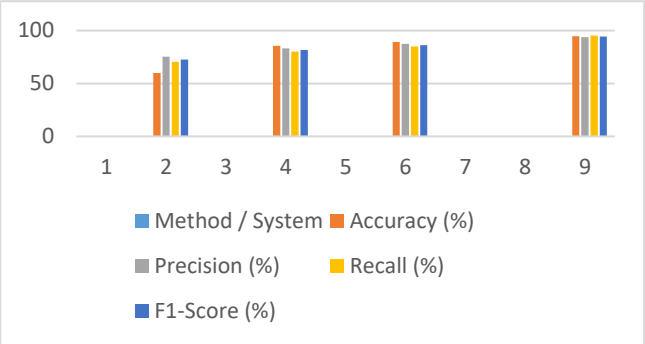


Fig: Comparison Graph

Parameter Group	Description	Unit
Environmental Conditions	Ambient temperature, relative humidity, toxic gas concentration, and overall air quality monitoring inside sewage environment	°C, %, ppm, Index
Physiological Parameters	Heart rate and oxygen saturation (SpO_2) monitoring to assess worker health status	bpm, %
System & Alert Status	Alert triggering and LoRa communication status for real-time monitoring and warnings	Binary (0/1)

Table 1 Parameter Table

An Independent Samples t-test will be used to compare the performance of a system under different conditions, including safe vs. hazardous environments, or manual vs. automated wearable monitoring. A paired samples t-test will also be conducted to compare the readings of the sensors prior to the installation of the wearable monitoring system and after to measure improvements in response times and accuracy of hazard detection. The t-statistic, degrees of freedom, p-value, mean difference, standard error of the difference, and the 95% confidence intervals will all be reported as part of the analysis. A significance level of $p < 0.05$ will indicate whether or not the differences found were statistically significant; i.e. whether the differences were statistically different enough to conclude that the proposed wearable monitoring system affords improved methods of early detection of toxic gases, environmental hazards, and workers' health risks relative to more traditional or manual methods of monitoring. This statistical analysis thus serves to establish that the proposed system is a real-time monitoring system that is both reliable and accurate for monitoring of the environment, sewerage. Thus, the proposed system represents an effective means of reducing hazards to workers in the sewerage industry, providing for timely intervention, and thus improved safety for employees working in the sewerage industry.

V. RESULTS

This project will be using an Arduino board (also known as the brain of the project) which will be connected to a GSM (Global System for Mobile Communications) module to notify the user with a message when it receives any input from the sensor modules as designed for the project. The web application will serve as a converter between the end user's device and the microcontroller. Both Wi-Fi modules and LCDs are used for this project. Arduino NANO has an 8-bit ATmega328P microcontroller (MC), but the ATmega328P doesn't stand alone, there are also other components that make up its functionality such as a crystal oscillator, serial ports to communicate with other devices, regulator circuits to supply voltage to the microcontroller, etc. Other features of the Arduino NANO include 14 digital input/output pins (of these 6 can be used as Pulse Width Modulation (PWM) outputs), 6 analogue input pins, a connection for a USB cable, an input for the power supply barrel jack, a header for the In Circuit Serial Programming (ICSP) connection, and a reset button. The ESP8266, a Wi-Fi microcontroller developed by Espressif Systems, is a fully integrated Wi-Fi solution used for connecting existing microcontrollers with the internet. The ESP8266 is a small, low-power, 32-bit MCU equipped with an industry-leading

Method / System	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Traditional Manual Monitoring	78.5	75.2	70.4	72.7
Wired Sensor-Based Monitoring	85.6	83.1	80.2	81.6
Existing Wireless Monitoring System	89.3	87.5	85.1	86.3
Proposed IoT-Based Sewage Monitoring System	94.8	93.6	95.2	94.4

Table 2 Comparison Table

The ESP8266 provides a complete TCP/IP stack and supports the IEEE802.11 b, g, and n wireless standards. By using the ESP8266 as a networking module in conjunction with an existing device, users can create their own wireless network or connect multiple devices directly to the Internet. In addition, the ESP8266 provides very low-cost wireless systems-on-chip (SoCs) and supports many other components included in a wireless system. An ESP8266 module allows for a fully integrated solution with minimal PCB footprint.

VI.CONCLUSION

This article discusses a power-efficient long communication distance based quality monitoring system based on IoT technology and a decentralized system of automated quality monitoring. This will address the limitations of traditional centralized quality monitoring systems. Our unique proposed quality monitoring system combines decentralized management of data, as well as creating trusted and tamper-resistant mechanisms. With these, we are able to guarantee the integrity, transparency, and reliability of data being produced within the distributed network of sensing nodes. By removing any single point of failure within the system, our decentralized architecture provides greater resiliency and trustworthiness than traditional central-based solutions, making it ideal for deploying large-scale and other remote quality monitoring systems. Our proposed solution utilizes the LoRaWAN communication protocol for effective communication, which was created to support high range, low-power wide-area network communication between IoT devices. LoRaWAN allows long-range transmission of data between networked devices with minimal energy usage.

VII. FUTURE WORK

There is considerable potential to enhance the performance, reliability, and scalability of the proposed Arduino-based ESP8266 GSM wearable monitoring system for sewage workers. A primary area of future expansion includes the integration of cloud platforms and mobile apps to allow real-time visualisation of data, long-term data storage, and remote monitoring by safety officers and management personnel. Furthermore, the ability to perform predictive analytics on environmental and physiological trends will provide support for proactive safety-based decisions. Future iterations of the system should include additional sensors such as state-of-the-art multi-gas sensors, SpO2 blood oxygen saturation sensors, body temperature sensors, and motion-based fall detection modules. This would provide a more complete monitoring solution for workers' health and hazardous sewage conditions. By including machine learning and/or smart decision supporting algorithms, detection of hazards can be made more accurate, fewer false alarms issued, and alert thresholds adjusted based on an individual worker's condition and historical data. Optimising energy efficiency in future iterations of the proposed system would involve the implementation of low-power hardware optimising techniques, advanced power management methodologies, and will extend the duration between battery replacements during field operations. In addition, utilising LPWAN (Low Power Wide Area Network) technology, such as LoRa or NB-IoT, either independently or in conjunction with Wi-Fi and GSM will dramatically improve the reliability of communicating from the underground and confined sewage environments that sewage workers may find themselves working. Ergonomics and miniaturization of hardware components within the design of a wearable device address both user comfort and acceptance when the device is worn for long periods of time. These developments will allow the proposed system to develop into a sophisticated, smart, and

scalable safety solution that improves employees' occupational health, reduces emergency response times, and allows for continuous safety protection for employees working within dangerous environments related to the sewage industry.

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